

Machine Learning Fundamentals

A Structured Course Document for RAG Evaluation

Chapter 1: Introduction to Machine Learning

Machine Learning (ML) is a branch of artificial intelligence that enables computer systems to learn patterns from data and make predictions or decisions without being explicitly programmed for every possible situation.

A traditional programming system generally follows:

Input + Rules → Output

In machine learning, the process is different:

Input + Expected Output → Learning Algorithm → Model

After training, the model can process previously unseen input and generate predictions.

For example, suppose a company wants to predict whether an email is spam. Instead of manually creating thousands of rules, a machine learning model can learn from historical emails labeled as either "spam" or "not spam."

1.1 Machine Learning Components

A typical machine learning system contains:

- Dataset
- Features
- Target variable
- Learning algorithm
- Model
- Training process
- Evaluation process
- Prediction process

A **feature** is an input variable used by a model.

A **target variable** is the variable that the model attempts to predict.

For example, in house-price prediction:

- House area = feature
- Number of bedrooms = feature
- Location = feature

- House price = target
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Chapter 2: Types of Machine Learning

Machine learning is commonly divided into three major categories:

1. Supervised Learning
2. Unsupervised Learning
3. Reinforcement Learning

2.1 Supervised Learning

In supervised learning, the training dataset contains both input features and known target values.

For example:

Hours Studied	Attendance	Exam Result
2	65%	Fail
4	72%	Fail
6	80%	Pass
8	91%	Pass

The model learns the relationship between hours studied, attendance, and exam result.

Supervised learning includes:

- Regression
- Classification

Regression

Regression predicts continuous numerical values.

Examples:

- House price
- Temperature
- Salary
- Sales revenue

Classification

Classification predicts categories or classes.

Examples:

- Spam / Not Spam
- Disease / No Disease

- Fraud / Not Fraud
 - Pass / Fail
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2.2 Unsupervised Learning

In unsupervised learning, the dataset does not contain labeled target values.

The algorithm attempts to discover hidden structures within the data.

Common unsupervised learning techniques include:

- K-Means clustering
- Hierarchical clustering
- Principal Component Analysis

For example, a retailer could use clustering to divide customers into groups based on purchasing behavior.

2.3 Reinforcement Learning

Reinforcement learning involves an agent interacting with an environment.

The agent performs actions and receives rewards or penalties.

The objective is to learn a strategy that maximizes cumulative reward.

Important reinforcement-learning concepts include:

- Agent
- Environment
- State
- Action
- Reward
- Policy

Reinforcement learning is commonly used in robotics, games, and autonomous decision-making systems.

Chapter 3: Dataset and Feature Engineering

A dataset is a collection of observations used for analysis or model training.

A dataset can contain:

- Rows representing observations
- Columns representing variables

For example, a student dataset may contain 5,000 rows and 12 columns.

3.1 Features

Features represent information provided to a model.

Suppose a dataset contains:

- Age
- Income
- Education
- Work experience

These variables can be used as features for predicting loan approval.

3.2 Target

The target is the output that the model attempts to predict.

For a loan-approval system:

Features: income, age, credit score, employment status

Target: loan approval

3.3 Feature Engineering

Feature engineering involves creating, transforming, or selecting features to improve model performance.

For example, if a dataset contains:

- Birth year
- Current year

a new feature called **Age** can be created.

Another example is converting a timestamp into:

- Hour
- Day
- Month
- Day of week

Good feature engineering can sometimes improve model performance more than changing the machine learning algorithm.

Chapter 4: Data Preprocessing

Raw datasets frequently contain missing values, duplicate records, inconsistent formats, and categorical variables.

Data preprocessing prepares the dataset for machine learning.

4.1 Missing Values

Suppose the following ages are recorded:

21, 22, 24, NULL, 27

The missing value can potentially be replaced using:

- Mean
- Median
- Mode
- A model-based estimate

For numerical data with extreme outliers, the median is often more robust than the mean.

4.2 Encoding Categorical Variables

Machine learning algorithms generally require numerical representations.

Suppose a gender column contains:

Male, Female, Female, Male

One-hot encoding can convert this information into numerical columns.

4.3 Feature Scaling

Two common scaling techniques are:

Standardization

Standardization transforms a feature using:

$$z = (x - \mu) / \sigma$$

where:

- x = original value
- μ = mean
- σ = standard deviation

Min-Max Scaling

Min-max scaling transforms values into a specified range, commonly 0 to 1:

$$x' = (x - x_{\min}) / (x_{\max} - x_{\min})$$

Scaling is particularly important for algorithms such as KNN and SVM.

Chapter 5: Linear Regression

Linear regression is a supervised learning algorithm used to predict continuous numerical values.

The basic equation is:

$$y = \beta_0 + \beta_1 x$$

where:

- y = predicted value
- β_0 = intercept
- β_1 = coefficient
- x = input feature

For multiple features:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

Example

Suppose a simple model predicts salary based on years of experience:

$$\text{Salary} = 30,000 + 8,000 \times \text{Experience}$$

For an employee with 5 years of experience:

$$\text{Salary} = 30,000 + 8,000(5)$$

$$\text{Salary} = 70,000$$

Linear regression typically minimizes the sum of squared errors.

Chapter 6: Logistic Regression

Despite its name, logistic regression is primarily used for classification.

It predicts a probability between 0 and 1.

The logistic function is:

$$P(y=1) = 1 / (1 + e^{-z})$$

where:

$$z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

A threshold can then be used to convert probability into a class.

For example:

- Probability $\geq 0.5 \rightarrow$ Class 1
- Probability $< 0.5 \rightarrow$ Class 0

Logistic regression is commonly used for:

- Binary classification
 - Medical diagnosis
 - Fraud detection
 - Customer churn prediction
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Chapter 7: Decision Trees

A decision tree predicts an output by repeatedly splitting data according to feature values.

A simplified tree might use:

Income > 50,000?

If yes, the model follows one branch.

If no, it follows another branch.

Decision trees can be used for both classification and regression.

Advantages

- Easy to interpret
- Handles numerical and categorical features
- Requires relatively little preprocessing
- Can model nonlinear relationships

Disadvantages

- Can easily overfit
- Small data changes can produce a very different tree
- Deep trees may have poor generalization

Two common measures used for classification trees are **Gini impurity** and **entropy**.

Chapter 8: Random Forest

Random Forest is an ensemble learning algorithm that combines multiple decision trees.

Instead of relying on a single tree, Random Forest creates many trees and combines their predictions.

For classification, the final prediction is commonly determined using majority voting.

For regression, predictions from individual trees are commonly averaged.

Example

Suppose five decision trees predict:

Cat, Cat, Dog, Cat, Dog

The Random Forest prediction is:

Cat

because three out of five trees predicted Cat.

Random Forest reduces the variance associated with individual decision trees and is generally more robust than a single decision tree.

Important Random Forest hyperparameters include:

- `n_estimators`
 - `max_depth`
 - `min_samples_split`
 - `min_samples_leaf`
 - `max_features`
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Chapter 9: Support Vector Machines

Support Vector Machine (SVM) is a supervised learning algorithm used primarily for classification.

The objective of a linear SVM is to find a decision boundary that separates classes while maximizing the margin between them.

The observations closest to the decision boundary are called **support vectors**.

SVM can also use kernel functions to model nonlinear relationships.

Common kernels include:

- Linear
- Polynomial
- Radial Basis Function (RBF)

- Sigmoid

The RBF kernel is particularly useful when class boundaries are nonlinear.

An important SVM hyperparameter is **C**, which controls the trade-off between achieving a large margin and correctly classifying training observations.

Chapter 10: K-Nearest Neighbors

K-Nearest Neighbors (KNN) predicts a sample based on nearby observations.

For classification, the algorithm examines the K closest training samples and typically selects the most common class.

For example, if:

K = 5

and the five nearest observations contain:

- 3 Red
- 2 Blue

the predicted class is **Red**.

KNN is called a lazy learning algorithm because it does not learn an explicit model during training. Most computation occurs when predictions are requested.

Distance metrics commonly include:

- Euclidean distance
- Manhattan distance
- Minkowski distance

Feature scaling is especially important for KNN because distance calculations are affected by feature magnitudes.

Chapter 11: Clustering

Clustering is an unsupervised learning technique that groups similar observations together.

11.1 K-Means

K-Means requires the user to specify the number of clusters, K.

The basic process is:

1. Select K initial centroids.
2. Assign each observation to its nearest centroid.

3. Recalculate the centroid of each cluster.
4. Repeat assignment and centroid calculation.
5. Stop when cluster assignments stabilize or a stopping criterion is reached.

For example, a company may use K-Means with **K = 4** to divide customers into four behavioral groups.

11.2 Choosing K

The elbow method can help identify a suitable number of clusters.

The method plots the within-cluster sum of squares against different K values.

A noticeable bend, or "elbow," can indicate a reasonable K.

Chapter 12: Model Evaluation

Model evaluation determines how well a trained model performs on unseen data.

The evaluation metric depends on the problem.

12.1 Classification Metrics

Important classification metrics include:

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC

Accuracy

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$$

Precision

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

Precision answers:

"Of all positive predictions, how many were actually positive?"

Recall

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

Recall answers:

"Of all actual positive cases, how many did the model identify?"

F1-Score

$$F1 = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

F1-score provides a balance between precision and recall.

Chapter 13: Overfitting and Underfitting

13.1 Overfitting

Overfitting occurs when a model learns the training data too closely, including noise and random patterns.

An overfitted model may have:

Training accuracy = 98%

Validation accuracy = 71%

This large difference suggests poor generalization.

Common methods for reducing overfitting include:

- Regularization
- Cross-validation
- Reducing model complexity
- Early stopping
- Increasing training data
- Feature selection
- Dropout for neural networks

13.2 Underfitting

Underfitting occurs when a model is too simple to capture important patterns.

An underfitted model may perform poorly on both training and validation datasets.

Chapter 14: Regularization

Regularization adds a penalty to the objective function to discourage overly complex models.

Two common forms are:

L1 Regularization

L1 adds a penalty based on the absolute values of coefficients.

$$\text{Loss} = \text{Original Loss} + \lambda \sum |\beta_i|$$

L1 regularization can force some coefficients to exactly zero, making it useful for feature selection.

L2 Regularization

L2 adds a penalty based on squared coefficients.

$$\text{Loss} = \text{Original Loss} + \lambda \sum \beta_i^2$$

L2 generally shrinks coefficients toward zero without necessarily making them exactly zero.

Chapter 15: Cross-Validation

Cross-validation evaluates how well a model generalizes.

In **5-fold cross-validation**, the dataset is divided into five approximately equal sections.

The model is trained and evaluated five times.

During each iteration:

- Four folds are used for training.
- One fold is used for validation.

Each fold becomes the validation set once.

The final performance can be calculated as the average of the five validation scores.

Cross-validation helps reduce dependence on a single train-validation split.

Chapter 16: Feature Selection

Feature selection identifies the most useful variables for a machine learning model.

Benefits include:

- Reduced computational cost
- Reduced noise
- Lower risk of overfitting
- Improved interpretability
- Potentially improved model performance

Common feature-selection approaches include:

Filter Methods

Features are selected using statistical characteristics.

Examples:

- Correlation
- Chi-square test
- Mutual information

Wrapper Methods

Different feature subsets are evaluated using a machine learning model.

Embedded Methods

Feature selection occurs during model training.

Examples include L1 regularization and tree-based feature importance.

Chapter 17: Ensemble Learning

Ensemble learning combines multiple models to improve prediction performance.

Two important ensemble strategies are:

Bagging

Bagging trains models independently on different samples of the training data.

Random Forest is an example of bagging.

Boosting

Boosting trains models sequentially, with later models focusing more on observations that previous models handled poorly.

Popular boosting algorithms include:

- AdaBoost
- Gradient Boosting
- XGBoost
- LightGBM
- CatBoost

Bagging primarily helps reduce variance, while boosting can reduce bias and improve predictive performance when properly tuned.

Chapter 18: Hyperparameter Tuning

Hyperparameters are settings selected before or during model training rather than learned directly from the training data.

Examples:

For Random Forest:

- Number of trees
- Maximum tree depth
- Maximum features

For KNN:

- Number of neighbors
- Distance metric

For SVM:

- C
- Kernel
- Gamma

Common tuning approaches include:

Grid Search

Grid Search evaluates predefined combinations of hyperparameters.

Random Search

Random Search samples hyperparameter combinations from specified distributions.

Random Search can be more efficient than exhaustive Grid Search when only a few hyperparameters strongly influence performance.

Chapter 19: Machine Learning Pipeline

A typical machine learning pipeline can contain the following stages:

Data Collection → Data Cleaning → Feature Engineering → Data Splitting → Preprocessing → Model Training → Hyperparameter Tuning → Evaluation → Deployment → Monitoring

19.1 Train-Test Split

A dataset can be divided into training and testing subsets.

A common split is:

80% training / 20% testing

The training set is used to learn model parameters.

The test set is reserved for final evaluation.

The test set should not influence model selection or hyperparameter tuning.

Chapter 20: Practical Case Study — Student Performance Prediction

Consider a university that wants to predict whether a student will pass an examination.

The dataset contains **2,000 student records**.

Each record contains:

- Study hours
- Attendance percentage
- Assignment score
- Previous GPA
- Number of missed classes
- Final examination result

The target variable is:

Pass = 1

Fail = 0

20.1 Data Split

The dataset is divided into:

- 1,600 training records
- 400 testing records

20.2 Candidate Models

Three models are trained:

Model	Accuracy
Logistic Regression	82%
Decision Tree	78%
Random Forest	87%

Random Forest produces the highest test accuracy of **87%**.

However, accuracy alone is not necessarily sufficient. The university should also examine precision, recall, and F1-score.

20.3 Important Finding

Suppose the Random Forest model achieves:

- Accuracy = 87%
- Precision = 84%
- Recall = 91%
- F1-score = 87%

The high recall indicates that the model identifies a large proportion of students who actually pass.

Chapter 21: Important Formulas

Mean

$$\text{Mean} = \Sigma x / n$$

Standard Deviation

$$\sigma = \sqrt{[\Sigma(x - \mu)^2 / n]}$$

Linear Regression

$$y = \beta_0 + \beta_1 x$$

Logistic Function

$$P(y=1) = 1 / (1 + e^{-z})$$

Accuracy

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

Precision

$$\text{Precision} = TP / (TP + FP)$$

Recall

$$\text{Recall} = TP / (TP + FN)$$

F1-Score

$$F1 = 2PR / (P + R)$$

Min-Max Scaling

$$x' = (x - x_{\min}) / (x_{\max} - x_{\min})$$

Standardization

$$z = (x - \mu) / \sigma$$

Chapter 22: Glossary

Algorithm: A procedure used to solve a computational problem.

Classification: A supervised learning task where the target is a category.

Clustering: An unsupervised learning technique that groups similar observations.

Feature: An input variable used by a machine learning model.

Hyperparameter: A model configuration selected before or during training.

Inference: The process of using a trained model to generate predictions.

Model: A learned mathematical representation of patterns in data.

Overfitting: When a model performs very well on training data but poorly on unseen data.

Precision: The proportion of positive predictions that are actually positive.

Recall: The proportion of actual positive cases correctly identified.

Regression: A supervised learning task that predicts continuous numerical values.

Regularization: A technique that penalizes model complexity.

Target: The variable that the model attempts to predict.

Training Data: Data used to learn model parameters.

Validation Data: Data used to evaluate models during development.

Test Data: Data reserved for final evaluation.
